

# Contents

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## 1 A proof that $\pi$ (and in fact $\pi^2$ ) is irrational

**Theorem.**  $\pi^2$  (and therefore  $\pi$ ) is irrational (As if  $\pi$  were rational  $\pi^2$  would be too). Because rational numbers are exactly those whose decimals repeat (proven in Level 2),  $\pi$ 's decimals never repeat.

*Proof. Step 1.* We need to set everything up

Start by assuming  $\pi^2 = \frac{a}{b}$ , we aim to derive a contradiction. Define the sequence of integrals  $I_n$  as follows:

1.  $I_0 = 2$
2.  $I_1 = 4b$
3.  $I_n = \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx$  if  $n \geq 2$

We can bound  $I_n$  as follows: It is non-negative since everything in the integral is non-negative and so is the coefficient, and  $x^n(\pi - x)^n$  is bounded above by  $\pi^n \pi^n = \pi^{2n}$  whenever  $0 \leq x \leq \pi$ , and therefore the integral (dropping the factor in front for now) can be bounded above by  $\int_0^\pi \pi^{2n} \sin(x) dx = 2\pi^{2n}$ . Therefore, if we consider  $I_n$ , we get

$$0 \leq I_n \leq 2 \frac{(\pi^2 b)^n}{n!}$$

Note that when, for example,  $n > 2\pi^2 b$ , the terms  $2 \frac{(\pi^2 b)^n}{n!}$  will be at most half of the previous term (since the numerator will be multiplied by  $\pi^2 b$  and the denominator by more than twice that), therefore these terms eventually tend to 0. Since  $I_n$  is bounded above by these terms and never negative,  $I_n$  also tends to 0.

The idea of the proof will be to show that if  $\pi^2 = \frac{a}{b}$  then  $I_n$  is an integer for all  $n$ , which is a problem since  $I_n$  is always strictly between 0 and something that tends to 0 (and therefore is eventually less than 1), so  $I_n$  can never be an integer and this will be our contradiction.

Now we will do a classic A level further maths problem where we show from algebra that for  $n$  at least 2,  $I_n = 2b(2n-1)I_{n-1} - abI_{n-2}$ , which would mean that  $I_n$  is always an integer (since  $a$ ,  $b$ ,  $n$ , and  $I_0, I_1$  are all integers) so we will be done.

**Step 2.** We need to verify what happens if  $n$  is 0 or 1. It turns out that the formula  $I_n = \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx$  holds for  $n=0$  and  $n=1$ . The proof is as follows.

**Case 1.** For  $n = 0$ , the integral simplifies to

$$\frac{b^0}{0!} \int_0^\pi x^0 (\pi - x)^0 \sin(x) dx = 1$$

Since  $0! = 1$  (If I haven't mentioned this, it's because  $2! = \frac{3!}{3}$ ,  $1! = \frac{2!}{2}$ , and  $0! = \frac{1!}{1} = 1$ )

**Case 2.** For  $n = 1$ , the integral simplifies to

$$b \int_0^\pi x (\pi - x) \sin(x) dx = \pi b \int_0^\pi x \sin(x) dx - b \int_0^\pi x^2 \sin(x) dx$$

Integration by parts on the first integral setting

$$u = x, \frac{dv}{dx} = \sin(x) \text{ so } \frac{du}{dx} = 1, v = -\cos(x)$$

gives that it is exactly equal to

$$-[x \cos(x)]_0^\pi - \int_0^\pi -\cos(x)dx = -(-\pi - 0) - 0 = \pi$$

Also,  $\int_0^\pi x^2 \sin(x)dx$  can be integrated by parts if we set

$$u = x^2, dv = \sin(x)dx \text{ so } du = 2x dx, v = -\cos(x)$$

meaning that the integral is then equal to

$$-[x^2 \cos(x)]_0^\pi - \int_0^\pi -2x \cos(x)dx = -(-\pi^2 - 0) + \int_0^\pi 2x \cos(x)dx = \pi^2 + \int_0^\pi 2x \cos(x)dx$$

Integrating the second integral by parts, setting

$$u = 2x, \frac{dv}{dx} = \cos(x) \text{ so } \frac{du}{dx} = 2, v = \sin(x)$$

we get that

$$\int_0^\pi 2x \cos(x)dx = [2x \sin(x)]_0^\pi - \int_0^\pi 2 \sin(x)dx = (0 - 0) - 4 = -4$$

Therefore we conclude

$$\int_0^\pi x^2 \sin(x)dx = \pi^2 - 4$$

Now  $I_1 = b\pi(\pi) - b(\pi^2 - 4) = 4b$ , as required.

**Step 3.** We will prove the recurrence relation  $I_n = 2b(2n - 1)I_{n-1} - abI_{n-2}$  for  $n$  at least 2. As discussed, this would establish  $I_n$  is always an integer but it is positive and tends to 0 which would give a contradiction. This is not conceptually difficult, just a very long calculus exercise.

Consider  $\int_0^\pi x^n(\pi - x)^n \sin(x)dx$ . We will do this by parts. Set

$$u = x^n(\pi - x)^n, \frac{dv}{dx} = \sin(x)$$

By the product and chain rules,

$$\frac{du}{dx} = nx^{n-1}(\pi - x)^n - nx^n(\pi - x)^{n-1} = x^{n-1}(\pi - x)^{n-1}[n(\pi - x) - nx] = nx^{n-1}(\pi - x)^{n-1}(\pi - 2x)$$

but  $u$  vanishes (is 0) at 0 and  $\pi$  and therefore the  $[uv]_0^\pi$  term from the integration by parts formula does too. Therefore the integration by parts formula gives (since  $v = -\cos(x)$ )

$$\int_0^\pi x^n(\pi - x)^n \sin(x)dx = n \int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x) \cos(x)dx$$

Now we want to integrate this by parts again so we need to work out the derivative of  $x^{n-1}(\pi - x)^{n-1}(\pi - 2x)$ . By the product rule on the last term, this is equal to

$$-2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x) \frac{d}{dx} [x^{n-1}(\pi - x)^{n-1}]$$

but this last derivative is just the  $n - 1$  version of something whose derivative we already know. Therefore the original derivative we wanted, ie the derivative of  $x^{n-1}(\pi - x)^{n-1}(\pi - 2x)$  is

$$-2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)^2(n - 1)x^{n-2}(\pi - x)^{n-2}$$

Therefore for the integral

$$\int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x) \cos(x)dx$$

we set

$$u = x^{n-1}(\pi - x)^{n-1}(\pi - 2x), \frac{dv}{dx} = \cos(x)$$

so now we have

$$v = \sin(x), \frac{du}{dx} = -2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2}$$

This time  $v$  vanishes at 0 and  $\pi$  so again the  $uv$  term vanishes and we reduce to

$$-\int_0^\pi v \frac{du}{dx} dx = \int_0^\pi \left[ 2x^{n-1}(\pi - x)^{n-1} - (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx$$

By some algebra, this is

$$\int_0^\pi \left[ 2x^{n-1}(\pi - x)^{n-1} - (\pi^2 - 4x(\pi - x))(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx$$

This is therefore equal to

$$\int_0^\pi \left[ (2 + 4(n-1))x^{n-1}(\pi - x)^{n-1} - (\pi^2)(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx$$

Putting all this together

$$\begin{aligned} I_n &= \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx = \frac{b^n}{n!} n \int_0^\pi x^{n-1} (\pi - x)^{n-1} (\pi - 2x) \cos(x) dx \\ &= \frac{b^n}{(n-1)!} \left[ \int_0^\pi \left[ 2 + 4(n-1) \right] x^{n-1} (\pi - x)^{n-1} - (\pi^2)(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx \\ &= b \frac{b^{n-1}}{(n-1)!} \int_0^\pi \left[ (2 + 4(n-1))x^{n-1}(\pi - x)^{n-1} \right] \sin(x) dx - \frac{b^2 \pi^2}{n-1} \frac{b^{n-2}}{(n-2)!} \left[ \int_0^\pi \left[ (n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx \right] \\ &= b I_{n-1} (2 + 4(n-1)) - b^2 \pi^2 \frac{b^{n-2}}{(n-2)!} \left[ \int_0^\pi \left[ x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx \right] \\ &= 2b(2n-1) I_{n-1} - ab I_{n-2} \end{aligned}$$

where we have used the fact that  $\pi^2 = \frac{a}{b}$  (We had to use it at some point otherwise we know something has gone wrong). This completes the proof of step 3.

□

## 2 A proof that $\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$

**Theorem.**  $\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$

*Proof.* Recall De Moivre's theorem, which says that  $(\cos(\theta) + i\sin(\theta))^n = \cos(n\theta) + i\sin(n\theta)$ . Now we have that

$$\begin{aligned} \frac{(\cot(\theta) + i)^{2n+1} - (\cot(\theta) - i)^{2n+1}}{2i} &= \frac{(\cos(\theta) + i\sin(\theta))^{2n+1} - (\cos(\theta) - i\sin(\theta))^{2n+1}}{2i(\sin^{2n+1}(\theta))} \\ &= \frac{\cos((2n+1)\theta) + i\sin((2n+1)\theta) - \cos((2n+1)\theta) + i\sin((2n+1)\theta)}{2i(\sin^{2n+1}(\theta))} = \frac{\sin((2n+1)\theta)}{\sin^{2n+1}(\theta)}. \end{aligned}$$

Now let's suppose that  $\theta = \frac{m\pi}{2n+1}$  for some integer  $m$  that is between 1 and  $n$  inclusive. Then we know that  $\sin((2n+1)\theta) = 0$  and therefore  $\frac{(\cot(\theta)+i)^{2n+1} - (\cot(\theta)-i)^{2n+1}}{2i} = 0$ . Now I will use the binomial theorem to expand  $(\cot(\theta) + i)^{2n+1} - (\cot(\theta) - i)^{2n+1}$ , which we know must be equal to 0. We will get

$$0 = \sum_{r=1}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) i^{2n+1-r} - \sum_{r=0}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) (-i)^{2n+1-r} = \sum_{r=1}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) \left[ i^{2n+1-r} - (-i)^{2n+1-r} \right].$$

When  $r$  is odd,  $2n + 1 - r$  is even, so the bracketed term will look like  $i^{2k} - (-i)^{2k}$  where  $k$  is an integer, but this equals  $(-1)^k - (-1)^k = 0$ . Therefore the only terms that survive are the terms where  $r$  is even, so we can replace  $r$  with  $s$  ranging from 1 to  $n$  where  $s$  is  $2r$ . We have

$$\sum_{s=0}^n \binom{2n+1}{2s} \cot^{2s}(\theta) \left[ i^{2(n-s)+1} - (-i)^{2(n-s)+1} \right]$$

Let's replace  $s$  with  $n-s$ , and note the identity  $\binom{2n+1}{2s} = \binom{2n+1}{2n+1-2s}$  (since we can choose  $2s$  things in the same number of ways we can exclude  $2s$  things). Let's call  $n-s$   $k$ , then we have

$$\sum_{k=0}^n \binom{2n+1}{2k+1} \cot^{2(n-k)}(\theta) \left[ i^{2k+1} - (-i)^{2k+1} \right]$$

This last term is  $i + i = 2i$  when  $k$  is even and  $-i - i = -2i$  when  $k$  is odd. Since the whole thing equals 0, we can divide it through by  $2i$  to get the following:

$$\sum_{k=0}^n \binom{2n+1}{2k+1} \cot^{2(n-k)}(\theta) (-1)^k = 0$$

If  $x = \cot^2(\theta)$  we can write the sum above as

$$\binom{2n+1}{1} x^n - \binom{2n+1}{3} x^{n-1} + \dots + (-1)^n = 0$$

Note that the roots of the polynomial above are exactly the  $n$  numbers  $\cot^2\left(\frac{m\pi}{2n+1}\right)$  for  $m$  an integer going from 1 to  $n$ . The same logic applies if  $m$  is an integer greater than  $n$  or less than 0, however this would give one of the same numbers – this can be argued from symmetry of the graph, or from the fact that it must be the case as  $\cot^2\left(\frac{m\pi}{2n+1}\right)$  for  $m$  from 1 to  $n$  is always distinct as  $\cot$  is increasing in that range, and an  $n$ -degree polynomial cannot have more than  $n$  roots (or else you could factor it more than  $n$  times by the factor theorem).

Now we know that the sum of roots of a polynomial is minus the second leading coefficient divided by the leading coefficient. Applying this to the polynomial above gives

$$\sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{\binom{2n+1}{3}}{\binom{2n+1}{1}} = \frac{\frac{(2n+1)!}{3!(2n-2)!}}{2n+1} = \frac{(2n)!}{6(2n-2)!} = \frac{(2n)(2n-1)}{6} = \frac{n(2n-1)}{3}$$

We now need two facts:

1. For  $\theta$  between 0 and  $\frac{\pi}{2}$ ,  $\sin(\theta) = \int_0^\theta \cos(x) dx$ . Since  $\cos(x)$  is always between 0 and 1 in that range, it means that  $\sin(\theta)$  is between 0 and  $\theta$
2. For  $\theta$  between 0 and  $\frac{\pi}{2}$ ,  $\tan(\theta) = \int_0^\theta \sec^2(x) dx$  (you can write  $\tan$  as  $\frac{\sin}{\cos}$  and differentiate it using the quotient rule and verify that it is  $\sec^2$ ). Since  $\cos(x)$  is always between 0 and 1 in that range, it means that  $\sec(\theta)$  is greater than 1 for all non-zero values in that range, and therefore  $\tan(\theta) > \theta$

So  $0 < \sin(\theta) < \theta < \tan(\theta)$  for  $\theta$  between 0 and  $\frac{\pi}{2}$ . Since in  $\sin(\theta) < \theta < \tan(\theta)$  they are all positive, taking the reciprocal of both sides flips the inequality, so  $\cot(\theta) < \frac{1}{\theta} < \csc(\theta)$ . We still have positive things, so we will square both sides to get  $\cot^2(\theta) < \frac{1}{\theta^2} < \csc^2(\theta)$ . However we can say something else, which is that  $\cot^2(\theta) < \frac{1}{\theta^2} < 1 + \cot^2(\theta)$ , as

$$\csc^2(\theta) = \frac{1}{\sin^2(\theta)} = \frac{\sin^2(\theta) + \cos^2(\theta)}{\sin^2(\theta)} = 1 + \cot^2(\theta)$$

Now

$$\sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right) < \sum_{m=1}^n \frac{(2n+1)^2}{m^2 \pi^2} < \sum_{m=1}^n 1 + \cot^2\left(\frac{m\pi}{2n+1}\right) = n + \sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right)$$

But we know

$$\sum_{m=1}^n \cot^2 \left( \frac{m\pi}{2n+1} \right) = \frac{n(2n-1)}{3}$$

Therefore

$$n + \sum_{m=1}^n \cot^2 \left( \frac{m\pi}{2n+1} \right) = \frac{n(2n+2)}{3}$$

Therefore we have the tight bound

$$\frac{n(2n-1)}{3} < \sum_{m=1}^n \frac{(2n+1)^2}{m^2 \pi^2} < \frac{n(2n+2)}{3}$$

for all positive integers n. Let's multiply through by  $\frac{\pi^2}{(2n+1)^2}$  to get that

$$\frac{n(2n-1)\pi^2}{3(2n+1)^2} < \sum_{m=1}^n \frac{1}{m^2} < \frac{n(2n+2)\pi^2}{3(2n+1)^2}$$

But notice that as n gets large,  $\frac{n(2n-1)}{3(2n+1)^2}$  and  $\frac{n(2n+2)}{3(2n+1)^2}$  will both approach  $\frac{1}{6}$ , as they are exactly  $\frac{1}{6}$  save for the +1 and +2 and -1 terms which contribute a factor on the order of  $\frac{1}{n}$ . Therefore the sum  $\sum_{m=1}^n \frac{1}{m^2}$  is bounded below and above by something that approaches  $\frac{\pi^2}{6}$ , and therefore the infinite sum is equal to exactly  $\frac{\pi^2}{6}$ .

□